

CT4101 – Machine Learning Portfolio Project

Predicting Single vs Multi-planet Systems from Host Star Characteristics

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Question:

Can a machine learning model predict what kind of system orbits a star using the host stars characteristics?

Dataset:

I will be using the Planetary Systems Composite Parameters (PSCompPars) [NASA Exoplanet Archive](#) developed by the NASA Exoplanet Science Institute located at the California Institute of Technology. License allows for public and academic use.

Target Variable:

Add a system type variable which is either single-planet (Number of Planets < 2) or multi-planet (Number of planets >= 2).

Main Input Features:

Stellar mass, Stellar Radius, Stellar Effective Temperature, Stellar Metallicity, Stellar Surface Gravity.

Dataset Quality and Scale:

There are more than 6000 confirmed planets in the exoplanet archive. All data is kept up to date by NASA astronomers. There are a lot of missing values but I have chosen my input features so that they have as little missing values as possible. There are 40 columns per row but I will remove around 35 of them.

Baseline, Candidate models and Evaluation metrics:

Most systems in the dataset only hold one known planet so I will use guessing single planet system for all rows as my baseline. My two candidate models will be a Decision Tree Classifier and a Random Forest Classifier. Given that this dataset has much more single-planet systems than multi-planet systems I will not be able to use overall accuracy. So instead I will calculate the accuracy individually for both single-planet and multi-planet systems to make sure that the model isn't just guessing single-planet every time. I will also use a 2x2 confusion matrix to see the full picture of how the model works.

Privacy, fairness, misuse and responsible-ML risks:

The data in this dataset is not a random sample of the systems around us because of selection bias. Astronomers find planets using methods like the transit method which measures the dip in light when a large planet orbits close to a bright nearby star. This means that the dataset only represents stars that are easier for our current telescopes to observe.

Repository: https://github.com/eunangallagher/exoplanet_system_classifier
[Christiansen et al. \(2025\)](#), the NASA Exoplanet Archive's published paper accepted by the *Planetary Science Journal* that provides an extensive overview of the archive's data, services and tools.